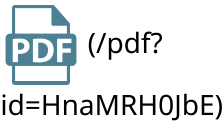




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Neural Dependency Region:
Improving Generalization through
Re-training Using Dependency
Region Extraction via Sensitivity
Analysis of Neural Operators



NeurIPS 2026 Conference Submission19582 Authors

03 May 2026 (modified: 27 May 2026)

NeurIPS 2026 Conference Submission

Conference, Senior Area Chairs, Area Chairs, Reviewers, Authors

Revisions (/revisions?id=HnaMRH0JbE)

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Abstract:
Neural operators are used to quickly solve partial differential equations by learning the maps between input and output functions. In many implementations, the input function is discretized into a finite number of sample points, or sensors. However, it remains unclear which input points contribute to the prediction. Consequently, the number and placement of sensors are determined empirically. Meanwhile, theoretical concepts such as the domain of dependence and the domain of influence, which are used to describe solution dependencies, have been developed in the field of partial differential equation theory. This research proposes a machine learning method to extract the domain of dependence from a trained neural operator. Furthermore, experiments demonstrate that placing sensors in the extracted domain and retraining can consistently improve test error while reducing the amount of data required.

Primary Area: AI/ML for physical sciences (e.g., physics, chemistry, Earth sciences, astronomy)

Secondary Area: General machine learning (e.g., core contributions in supervised and unsupervised methods)

Contribution Type: General: Most submissions will fall into this type.

Submission Number: 19582

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Official Review by Reviewer Lris 09 Jun 2026, 09:21 (modified: 09 Jun 2026, 09:31)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewer Lris Revisions (/revisions?id=1jCOHFKaML)

Summary:

The paper proposes a data-driven approach to estimate the input dependency regions of trained neural operators using Jacobian-based sensitivity analysis. By calculating the Jacobian energy of the output with respect to input sensors, the authors map out which regions most heavily influence the prediction analogous to the domain of dependence in partial differential equations (PDEs). They leverage this by applying a Top-k sensor selection strategy, where the model is retrained on only the most critical input points. This leads to improved generalization and reduced input dimensionality, validated across multiple PDEBench tasks and architectures.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths:

Theoretical Grounding: The connection drawn between Jacobian energy, Green’s functions, and variance contribution provides a robust mathematical justification for why the sensitivity maps work.

Empirical Consistency: The experimental setup is thorough, evaluating across diverse PDE types (hyperbolic, parabolic, elliptic) and architectures (DeepONet, Transformers, KANs), demonstrating consistent test error improvements.

Practical Utility: The Top-k selection and retraining loop turns a post-hoc interpretability tool into an actionable method for sensor dimensionality reduction.

Weaknesses:

Computational Overhead: The method requires a fully trained model to calculate the sensitivities before the input space can be reduced for retraining. This effectively doubles the training pipeline, which limits its utility in highly resource-constrained environments.

Architecture Variance: The performance gains are noticeably stronger for DeepONet architectures compared to Transformer-based models, suggesting the benefits might be partially tied to how specific models handle spatial encodings.

Quality: 3: good
Clarity: 4: excellent
Significance: 3: good
Originality: 3: good

Questions:

How does the computational overhead of computing the Jacobian-energy across the dataset compare to the training time saved by reducing the sensor inputs during retraining?

Is it feasible to estimate the dependency map dynamically during the initial training epochs (active learning) to bypass the need for a full pre-training cycle?

What is the intuition behind the diminishing returns observed in Transformer-based operators compared to DeepONets? Does the self-attention mechanism naturally mitigate the negative effects of overly dense, uninformative sensors?

Limitations:

Yes. The authors appropriately acknowledge the primary limitation in the conclusion: the method currently requires a pre-trained model with a baseline degree of accuracy to extract meaningful dependency maps.

Rating: 5: Accept: Technically solid paper, with high potential value on at least one sub-area of AI or moderate-to-high impact on more than one area of AI, with good-to-excellent evaluation, resources, reproducibility, and no unaddressed ethical considerations.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related

work.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:
N/A

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

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LLM Interaction by Reviewer Lris (👁 Vivek Kumar Mishra (/profile?id=~Vivek_Kumar_Mishra1))

📅 09 Jun 2026, 09:17 👁 Conference, AI Review Assistant, Reviewer Lris

Message:
what are strengths and weakness.

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LLM Interaction by AI Review Assistant 📅 09 Jun 2026, 09:17 (modified: 09 Jun 2026, 09:18)

👁 Conference, AI Review Assistant, Reviewer Lris 📄 Revisions (/revisions?id=2Va7PfyaML)

Message:
Here are **review-relevant strengths and weaknesses to inspect**, without turning them into a submitted review.

Potential strengths

- **Simple, intuitive method.** The core idea—use Jacobian energy $(\partial \hat{u}(y) / \partial u_0(\xi_j))^2$ to estimate which input sensors matter—is easy to understand and implement with autodiff.
- **Connects interpretability to action.** The paper does not only visualize sensitivity maps; it uses them for **Top-*k* sensor selection followed by retraining**, which makes the method practically testable.
- **PDE intuition is well motivated.** The discussion of domain of dependence/influence for hyperbolic, parabolic, and elliptic PDEs gives a natural conceptual basis for why input dependency regions should matter.
- **Architecture breadth.** The experiments claim results across DeepONet, Transformer/OFormer-style models, and KAN-based operators, which is stronger than demonstrating the method on one architecture only.
- **Some reproducibility details are provided.** The appendix includes data splits, sensor definitions, hyperparameters, metric definitions, and seed/trial counts.

Potential weaknesses / concerns to examine

- **Novelty may be limited.** The method is essentially gradient-based saliency plus Top-*k* feature/sensor selection and retraining. The paper should be judged on whether the operator-learning/PDE framing and empirical evidence are enough beyond existing saliency and feature-selection methods.
- **Baselines seem weak.** The main comparisons are only Top-*k* vs Uniform vs Random. Missing potentially important baselines include:
 - input-gradient selection without retraining,
 - integrated gradients / SmoothGrad,
 - learned sensor placement,
 - mutual-information or submodular sensor placement,
 - magnitude/variance-based feature selection,

- active learning or adaptive sensing baselines.
- **There are internal inconsistencies.**
 - The paper says the cross-model evaluation suggests the effect is “largely specific to DeepONet” around Sec. 5.4, but Table 5 appears to argue the opposite: gains are reported for all three model families. This may be a typo, but it affects interpretation.
 - Checklist says “**This paper does not include theoretical results**”, yet Sec. 6 and Appendix K present theoretical rationale and approximate optimality claims.
 - Table 5’s reported geometric mean gains look suspicious relative to Table 2. For example, DeepONet Uniform gains in Table 2 are mostly well above 1.0, but Table 5 reports Uniform geo-mean gain as 1.01.
- **Experimental reporting is incomplete.** Appendix H says full sweep tables with Top- k /Uniform/Random mean $\pm$ std are provided, but the uploaded text only contains a placeholder description, not the actual tables. Many appendix task tables report only Top- k and no standard deviations.
- **Statistical evidence is thin.** The paper reports mean $\pm$ std in some places, but:
 - model seeds are only 3,
 - Top- k seed appears to be 1,
 - no significance tests or confidence intervals are provided,
 - random baselines have 30 trials, but it is unclear how this interacts with model retraining variability.
- **The “data required” claim may be imprecise.** The method reduces the number of input sensors, but it still requires training a full-input model first, estimating sensitivities, and then retraining. So it reduces sensor/input dimensionality, not necessarily total training data or compute.
- **The theory is limited.** The Green’s-function argument is convincing for linear operators, but much weaker for nonlinear PDEs and trained neural operators. Appendix K depends on local linearization, diagonal/weakly correlated input covariance, and mean-imputation assumptions.
- **Possible circularity in sensor selection.** The method uses a trained full-input model to select sensors, then retrains. This is useful, but in realistic sensing problems one may not have full-input observations available initially.
- **Alignment metric may be questionable.** For advection with $c = 1$, $T = 1$, and periodic domain length $L = 1$, the upstream interval $[y - cT, y]$ may cover essentially the whole periodic domain depending on implementation. That could make the reference dependency map less informative.
- **Computational cost under-discussed.** The method may require many Jacobian computations over samples and query points. The paper mentions batching and approximations but does not provide concrete timing/memory costs in the paper itself.
- **Some claims may be overstated.** Phrases like “consistently improves” and “robust foundation” may be stronger than what is justified, especially given incomplete full-sweep reporting and limited baselines.

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LLM Interaction by AI Review Assistant 📅 30 May 2026, 02:01

👁 Conference, AI Review Assistant, Reviewer Lris

Message:

I have access to the paper, and am ready to assist with the review as needed. Please let me know how you'd like to proceed!

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